Title: Explainable AI Framework for Employee Attrition Prediction

Abstract :

Employee attrition is one of the major challenges faced by organizations, resulting in increased recruitment costs, productivity loss, and difficulties in retaining skilled employees. Traditional machine learning approaches generally predict whether an employee is likely to leave the organization but often fail to provide transparent explanations for their predictions and offer limited insight into why a particular prediction was made. This project proposes **an Explainable AI Framework for Employee Attrition Prediction** that leverages machine learning models to identify employees who are at risk of leaving while providing interpretable explanations for the model's decisions.

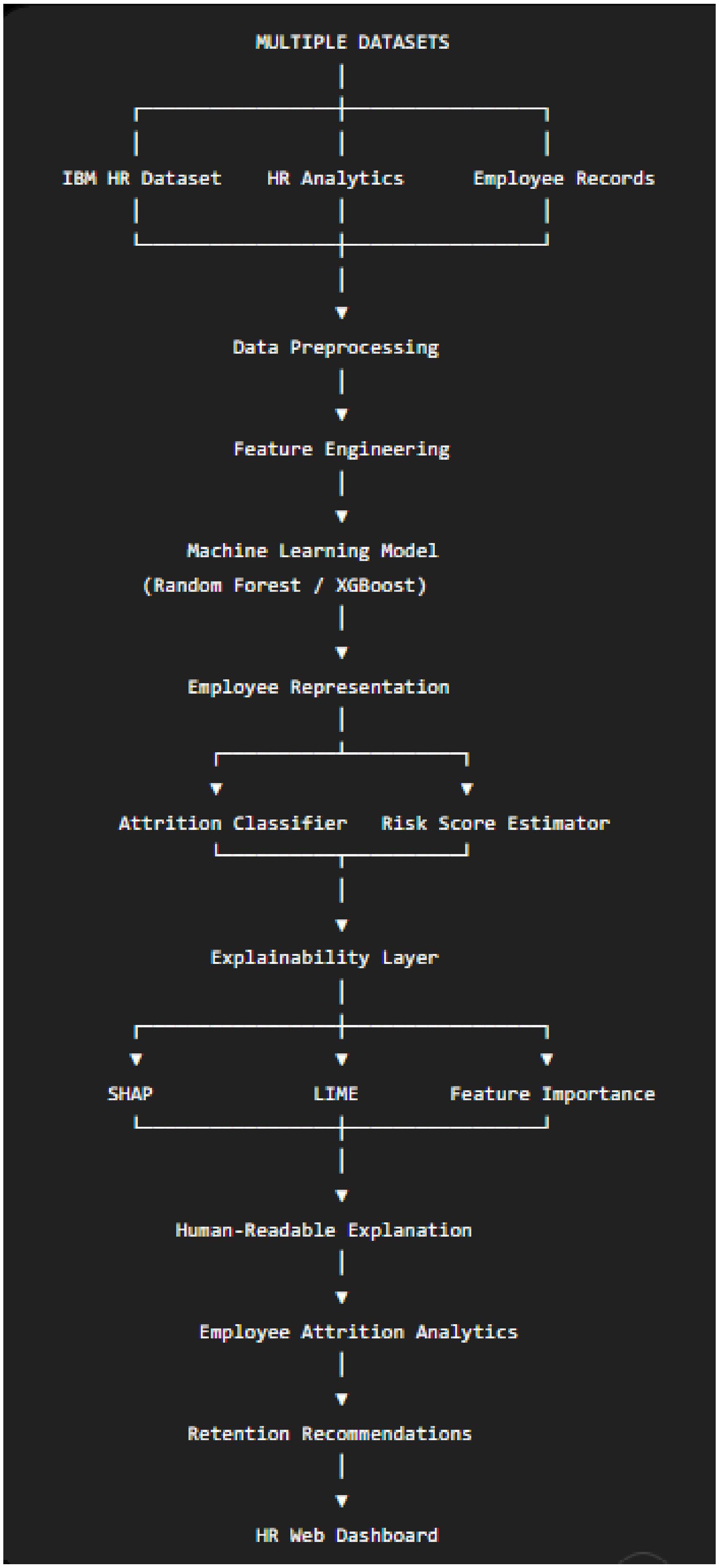
The proposed system uses supervised machine learning algorithms such as Random Forest, XGBoost, or Logistic Regression and trains them on benchmark employee attrition datasets such as the IBM HR Analytics Employee Attrition Dataset. The system performs data preprocessing, feature engineering, employee attrition classification, and prediction confidence estimation. To improve interpretability, explainable AI techniques such as SHAP, LIME, feature importance analysis, or permutation importance are incorporated to identify the employee attributes that contribute most significantly to the predicted attrition risk. The framework can analyze factors such as age, job role, monthly income, overtime, work-life balance, job satisfaction, years at the company, performance rating, promotion history, and distance from home, depending on the selected dataset and feature set.

The proposed approach can further analyze employee data at an aggregate level to identify departments with high attrition rates, recurring causes of employee turnover, emerging workforce trends, and areas requiring management attention. The performance of the machine learning model is evaluated using accuracy, precision, recall, F1-score, ROC-AUC score, and confusion matrix, while the explainability component is assessed based on the quality and consistency of generated explanations. The resulting system aims to provide not only accurate employee attrition prediction but also transparent and actionable insights that can support employee retention strategies, human resource planning, workforce optimization, and organizational decision-making.

Datasets :

Dataset	URL Link
IBM HR Analytics Employee Attrition	https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset
HR Employee Attrition	https://www.kaggle.com/search?q=employee+attrition
Employee Attrition Prediction	https://archive.ics.uci.edu/

Approach :


MULTIPLE DATASETS
IBM HR Dataset
HR Analytics
Employee Records
Data Preprocessing
Feature Engineering
Machine Learning Model
(Random Forest / XGBoost)
Employee Representation
Attrition Classifier
Risk Score Estimator
Explainability Layer
SHAP
LIME
Feature Importance
Human-Readable Explanation
Employee Attrition Analytics
Retention Recommendations
HR Web Dashboard

Project Group Members

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